

$$\begin{aligned}
& \max_{x \in \{0,1\}^n} \sum_{i=1}^n w_i x_i \\
& \text{s.t.} \quad \sum_{i=1}^n a_{ji} x_i \leq b_j \quad \forall j \in [m]
\end{aligned} \tag{1}$$

$$x_i \in \{0,1\} \forall i \in [n] \tag{2}$$